

DMG-01 GAME BOY ZERO: MASTER WIRING BLUEPRINT

SECTION 1: Front Isolated Button Board Layout

All button contact pads utilize a **Common Ground** configuration. Wire one side of every conductive pad to a shared Ground plane (GND). Trace the isolated data pad of each button directly to the designated Raspberry Pi GPIO pin below.

Game Boy Button Pad	Target Pi Zero 2 W GPIO Pin	Physical Pi Header Pin
D-Pad UP	GPIO 12	Pin 32
D-Pad DOWN	GPIO 13	Pin 33
D-Pad LEFT	GPIO 16	Pin 36
D-Pad RIGHT	GPIO 19	Pin 35
Action Button A	GPIO 26	Pin 37
Action Button B	GPIO 20	Pin 38
Action Button X	GPIO 21	Pin 40
Action Button Y	GPIO 6	Pin 31
SELECT	GPIO 5	Pin 29

START	GPIO 0	Pin 27
COMMON GROUND	GND	Pin 30 (<i>or any Pi GND pin</i>)

SECTION 2: 2.6" ST7789 SPI Display Pin Mapping

The 2.6-inch 240x320 SPI screen connects via a surface-mount FPC ribbon slot connector footprint on your PCB canvas. Ensure your software configures the Pi's secondary SPI bus (SPI1) if your main SPI bus is dedicated to the cartridge reader.

Display Ribbon Pin / Signal	Pi Zero 2 W Designati on	Physical Pi Header Pin	Function / Purpose
VCC	3.3V Power	Pin 17	Core Logic Power (3.3V Input)
GND	Ground	Pin 25	Common System Ground
SCL / SCLK	GPIO 21 (SPI1 SCLK)	Pin 40 (<i>Shared</i>)	Master Clock Signal
SDA / MOSI	GPIO 20 (SPI1 MOSI)	Pin 38 (<i>Shared</i>)	Master Out / Data Input

RES / RESET	GPIO 25	Pin 22	Hardware Reset Line
DC / RS	GPIO 24	Pin 18	Data / Command Selection
CS / CE	GPIO 18 (SPI1 CE0)	Pin 12	Chip Enable Line
LEDA / BACKLIGHT	5V Power <i>(via 100Ω Resistor)</i>	Pin 2	Backlight LED Power Supply

SECTION 3: MicroSD SMT Cartridge Slot Passthrough

The surface-mount MicroSD socket (GCT MEM2075) sits inside the empty cartridge frame shell, routing power and data lines directly through the console's desoldered 32-pin edge jack using the main SPI bus (SPI0).

MicroSD Socket Pin	Signal Name	Target Pi Zero 2 W GPIO Pin	Physical Pi Pin
Pin 1 <i>(DAT2)</i>	NC / Pull- up	<i>Not Connected</i>	—
Pin 2 <i>(CD/DAT3)</i>	CS (Chip Select)	GPIO 8 (SPI0 CE0)	Pin 24

Pin 3 (<i>CMD</i>)	MOSI (Data In)	GPIO 10 (SPI0 MOSI)	Pin 19
Pin 4 (<i>VDD</i>)	3.3V Power	3.3V Regulated Output	Pin 1
Pin 5 (<i>CLK</i>)	SCK (Clock)	GPIO 11 (SPI0 SCLK)	Pin 23
Pin 6 (<i>VSS</i>)	GND	System Ground	Pin 6
Pin 7 (<i>DAT0</i>)	MISO (Data Out)	GPIO 9 (SPI0 MISO)	Pin 21
Pin 8 (<i>DAT1</i>)	NC / Pull- up	<i>Not Connected</i>	—

SECTION 4: Step-by-Step KiCad & Assembly Workflow

Step 1: Footprint Asset Checklist

Verify that your KiCad workspace has loaded these explicit native and community profiles before tracing:

- Controls Area:** `Button_Switch_Keyboard:Conductive_Rubber_Pad_DMG`
(Provides the physical surface contact wheel comb layers for the rubber conductive pads).
- Extra Buttons:** `Button_Switch_SMD:SW_SPST_OMRON_B3F-10xx` (If opting for physical SMT tactile micro-switches for your hidden rear or front X/Y buttons).
- Cartridge Profile:** `GumpyFunction_GameBoy_Cartridge_Template`
(Maintains physical circuit dimensions matching the console interior casing

tracks).

4. **Display Interface:** `Connector_FFC-FPC:Hirose_FH12-10S-0.5SH` (For routing the flexible display connector lines).

Step 2: Solder Paste Stencil Reflow

Because hand-soldering the 1.1mm gaps on the GCT MEM2075 or the 0.5mm FPC slots is highly prone to bridging terminals, execute a paste reflow:

1. Align the custom **Stainless Steel SMD Stencil** flat across your unpopulated manufactured PCB panel.
2. Apply a thin, even bead of **SMT Solder Paste** across the stencil cutouts using a squeegee blade or card.
3. Lift the stencil cleanly away, then gently drop your surface-mount MicroSD socket and FPC jack directly onto the tacky gray paste footprints using fine tweezers.
4. Set your **Hot Air Rework Station** to **320°C (608°F)** with low airflow. Hold the nozzle vertically above the components until you see the paste liquidity turn bright silver and snap onto the copper pads via surface tension.

Step 3: Software Configurations

To configure Raspberry Pi OS or RetroPie to automatically mount your custom hardware slot insertions at boot, map an overlay drive array within the system:

1. Open the file `/boot/firmware/config.txt` inside your terminal editor.
2. Inject this instruction line to register your target console edge signals as a functional storage module:

text

```
dtoverlay=spi-sdcard,overclock_50=1
```

Use code with caution.

3. Save the configurations and restart the Pi. The operating system will automatically populate and scan directories through your custom Game Boy cartridges exactly like an integrated external hard drive.